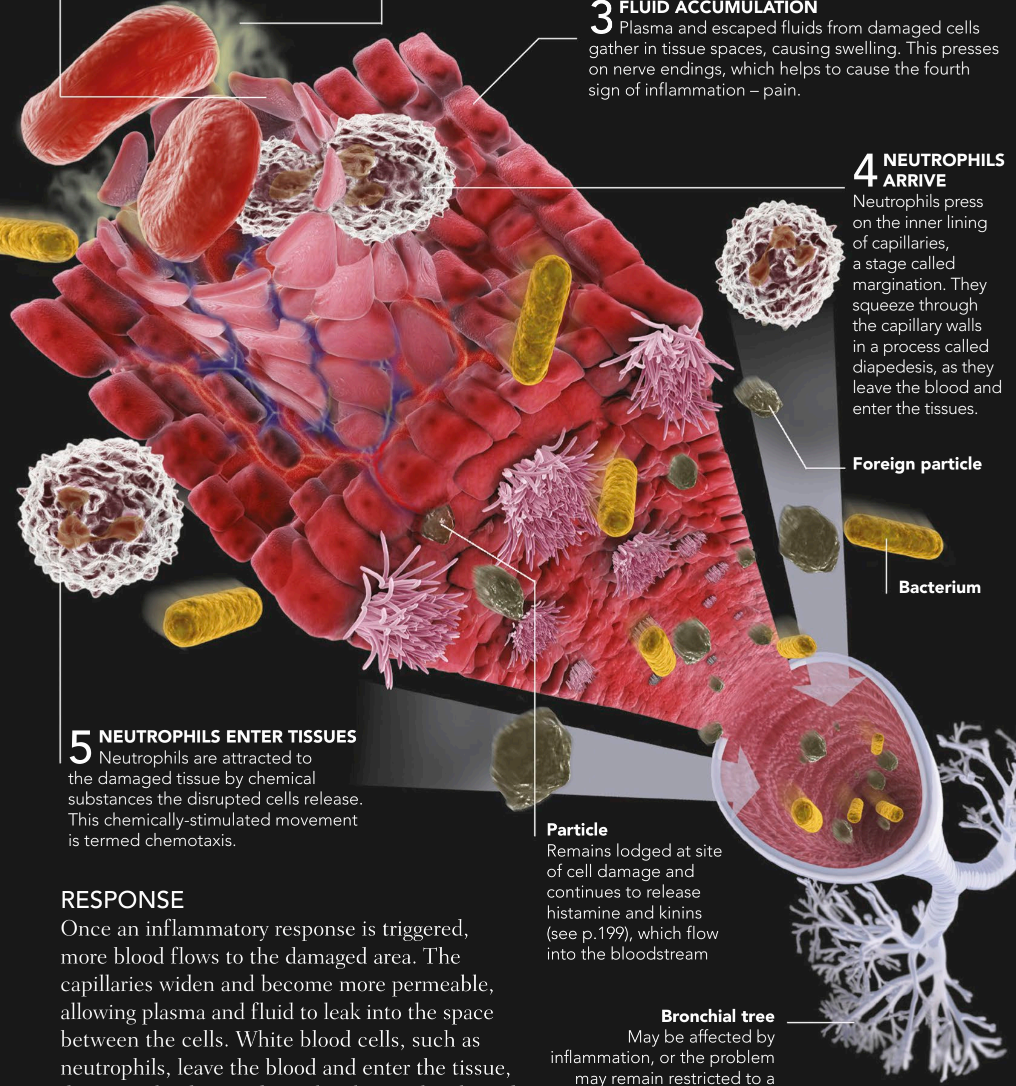
1 CAPILLARIES DILATE
Histamine stimulates widening of capillaries (vasodilation). As their walls stretch and become thinner, narrow gaps appear and make them more permeable to fluids.

2 FLUID LEAKAGE
Increased blood flow produces redness and heat. Plasma (blood's liquid component, pictured as yellow) leaks into the space between the cells, carrying various proteins such as fibrinogen, which helps blood to clot when the skin is broken.

3 FLUID ACCUMULATION
Plasma and escaped fluids from damaged cells gather in tissue spaces, causing swelling. This presses on nerve endings, which helps to cause the fourth sign of inflammation – pain.

4 NEUTROPHILS ARRIVE
Neutrophils press on the inner lining of capillaries, a stage called margination. They squeeze through the capillary walls in a process called diapedesis, as they leave the blood and enter the tissues.



5 NEUTROPHILS ENTER TISSUES
Neutrophils are attracted to the damaged tissue by chemical substances the disrupted cells release. This chemically-stimulated movement is termed chemotaxis.

Particle
Remains lodged at site of cell damage and continues to release histamine and kinins (see p.199), which flow into the bloodstream

Bronchial tree
May be affected by inflammation, or the problem may remain restricted to a patch of the trachea

RESPONSE

Once an inflammatory response is triggered, more blood flows to the damaged area. The capillaries widen and become more permeable, allowing plasma and fluid to leak into the space between the cells. White blood cells, such as neutrophils, leave the blood and enter the tissue, drawn to the damaged area by chemicals released from the disrupted cells.